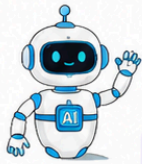


Subject Code: DI05016021

Subject Name: AI Product Design

Unit-1 Fundamentals of AI Products & Emerging AI Systems



UNIT 1: FUNDAMENTALS OF AI PRODUCTS & EMERGING AI SYSTEMS

1.1 INTRODUCTION TO AI PRODUCTS

An AI product is a software or digital solution that uses Artificial Intelligence to solve real problems and create value for users.

Examples:



Google Maps
(traffic prediction)



Netflix
(movie recommendation)



Spotify
(song recommendation)



ChatGPT
(text generation)

Key Points

- AI products learn from data.
- They make predictions, suggestions or decisions.
- They improve over time with more data and feedback.
- Goal: Solve user problems effectively and efficiently.

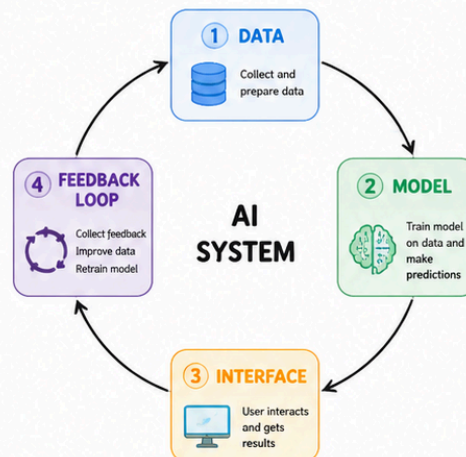
1.2 DIFFERENCE BETWEEN AI TOOL AND AI PRODUCT

Basis	AI Tool	AI Product
Meaning	 A single feature or model that performs a specific task.	A complete solution that uses AI to solve a user problem from end to end. 
Scope	Limited – does one thing.	Wide – includes many features and systems.
User Focus	Focus on the task or output.	Focus on user experience and value.
Example	Face recognition API	A smart attendance system that uses face recognition API, stores data, shows reports, sends alerts, etc.
Analogy	A camera (takes pictures) 	A mobile phone (camera + apps + storage + communication) 

1.3 COMPONENTS OF AI SYSTEMS

An AI system is made up of four main components that work together.

- 1 DATA**  Data is the information used to train and test the AI model. Good quality data leads to better results. Types of data: Text, Images, Numbers, Audio, Video, etc.
- 2 MODEL**  Model is the brain of the AI system. It learns patterns from data and makes predictions or decisions. Examples: Machine Learning models, Deep Learning models, LLMs (Large Language Models).
- 3 INTERFACE**  Interface is how the user interacts with the AI system. It can be a website, mobile app, chatbot, voice assistant, dashboard, etc.
- 4 FEEDBACK LOOP**  Feedback loop collects user responses and outcomes. This feedback is used to improve the data, retrain the model and make the system better over time.



In simple words: **Data** is the fuel, **Model** is the engine, **Interface** is the vehicle, and **Feedback loop** is the GPS that keeps the system on the right path and helps it improve!

Overview of Unit-1

1.1 Introduction to AI Products

What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a technology that enables computers to think, learn, and make decisions like humans.

AI can:

- Understand information
- Learn from data
- Solve problems
- Make predictions
- Generate content

What is an AI Product?

An AI Product is a software application or digital system that uses AI to solve real-world problems and help users.

Simple Definition

AI Product = AI Technology + User Interface + Useful Solution

Examples of AI Product

AI products are software applications or digital systems that use Artificial Intelligence (AI) to help users solve problems, make decisions, or perform tasks more efficiently. These products learn from data and improve their performance over time.

1. ChatGPT

ChatGPT is an AI-powered chatbot developed by OpenAI. It can understand natural language and generate human-like responses. Users can ask questions, write emails, generate code, summarize documents, and create content. ChatGPT uses Large Language Models (LLMs) to understand and generate text.

Purpose: Text generation, question answering, and conversational assistance.

2. Google Maps

Google Maps uses AI to provide navigation and route recommendations. It analyzes traffic conditions, road closures, and historical travel data to suggest the fastest route. The system continuously updates routes based on real-time information.

Purpose: Navigation and traffic prediction.

3. Netflix

[Netflix](#) uses AI to recommend movies and TV shows based on a user's viewing history, ratings, and preferences. The recommendation system helps users discover content they are likely to enjoy.

Purpose: Personalized content recommendation.

4. Spotify

[Spotify](#) uses AI algorithms to analyze listening habits and recommend songs, playlists, and podcasts. Features such as Discover Weekly provide personalized music suggestions to users.

Purpose: Music recommendation and personalization.

5. Amazon

[Amazon](#) uses AI to recommend products based on customers' browsing history, purchases, and interests. AI also helps in inventory management, demand forecasting, and customer support.

Purpose: Product recommendation and e-commerce optimization.

6. Google Translate

[Google Translate](#) uses AI and machine learning to translate text, speech, and images from one language to another. It supports many languages and continuously improves translation quality.

Purpose: Language translation.

7. YouTube

YouTube uses AI to recommend videos based on a user's watch history, search behavior, and interests. AI also helps detect inappropriate content and generate captions.

Purpose: Video recommendation and content management.

8. Smart Attendance System

A Smart Attendance System uses AI-based face recognition technology to automatically identify students or employees and record attendance. It eliminates manual attendance processes and improves accuracy.

Purpose: Automated attendance management.

Why Are AI Products Important?

AI products help users:

- ✓ Save time
- ✓ Increase productivity
- ✓ Make better decisions
- ✓ Automate repetitive tasks
- ✓ Improve user experience

Real-Life Example

Imagine you use Google Maps.

- You enter a destination.
- AI checks traffic data.
- AI finds the fastest route.
- AI updates the route if traffic changes.

Therefore, Google Maps is an AI Product because it solves a real-world problem using AI.

Characteristics of AI Products

Good AI products:

1. Learn from Data

They improve by analyzing large amounts of data.

2. Make Predictions

They predict what may happen next.

Example:

Netflix predicts which movie you may like.

3. Personalize Experience

Different users receive different suggestions.

Example:

Spotify recommends songs based on your listening history.

4. Improve Over Time

The more data they receive, the smarter they become.

1.2 Difference Between AI Tool and AI Product

An **AI Tool** is a technology or software component that performs a specific task using Artificial Intelligence. It is designed to solve only a particular problem. For example, a face recognition model can identify faces in images, but it cannot manage attendance records, generate reports, or communicate with users. Therefore, it is considered an AI Tool.

An **AI Product** is a complete solution that uses one or more AI tools to solve a real-world problem for users. It includes not only AI technology but also a user interface, database, security features, reporting systems, and other functionalities required by users. For example, a Smart Attendance System uses a face recognition model along with student records, attendance management, report generation, and notification services. Therefore, it is an AI Product.

The main focus of an AI Tool is performing a specific function, whereas the main focus of an AI Product is providing value to users by solving their problems. AI Tools are usually building blocks that developers use to create AI Products.

Consider the example of a car. The engine helps the car move, but it cannot transport people by itself. Similarly, an AI Tool performs a specific task. A complete car contains an engine, wheels, seats, brakes, and steering system working together to provide transportation. In the same way, an AI Product combines AI tools with many other components to provide a complete solution.

Example 1

A translation model that converts English text into Gujarati is an AI Tool because it performs only one task.

Google Translate is an AI Product because it provides a complete translation service through a user-friendly application.

Example 2

A recommendation algorithm is an AI Tool because it only suggests items based on user preferences.

Netflix is an AI Product because it uses recommendation algorithms along with video streaming, user profiles, subscriptions, and content management to provide a complete entertainment platform.

Key Difference

An AI Tool performs a single intelligent task, whereas an AI Product combines AI with software features and services to solve real-world problems for users.

AI Tool

An AI Tool performs a specific task.

It is usually a component or technology used to build larger systems.

AI Product

An AI Product is a complete solution that users directly use to solve problems.

Easy Example

AI Tool = Engine

An engine helps a car move.

But an engine alone cannot transport people.

AI Product = Car

A car contains:

- Engine
- Wheels
- Seats
- Steering
- Fuel system

Together they solve the transportation problem.

Therefore:

Engine = AI Tool

Car = AI Product

Another Example

AI Tool

Face Recognition Model

It only identifies faces.

AI Product

Smart Attendance System

It includes:

- Face recognition
- Student database
- Attendance records
- Reports
- Notifications

1.3 Components of AI Systems

An AI system is made up of several components that work together to provide intelligent solutions. Just as a car needs fuel, an engine, a steering wheel, and a driver to function properly, an AI system also requires different components to operate effectively.

The four main components of an AI system are:

1. Data
2. Model
3. Interface
4. Feedback Loop

These components continuously interact with each other to improve the performance of the AI system.

1.3.1 Data

What is Data?

Data is a collection of facts, observations, or information that an AI system uses to learn and make decisions.

Data is the foundation of every AI system. Without data, AI cannot learn anything.

Simple Definition

Data is the information used to train and test an AI system.

Just as students learn from books and examples, AI systems learn from data.

Examples of Data

- Text documents
- Images
- Audio recordings
- Videos
- Numbers and statistics

- Customer information
 - Sensor readings
-

Importance of Data

The quality of data directly affects the quality of AI results.

Good Data

- Accurate
- Complete
- Relevant
- Up-to-date

Produces better predictions and decisions.

Poor Data

- Incorrect
- Incomplete
- Outdated

Produces poor and unreliable results.

Real-Life Example

In a face recognition system:

- Thousands of face images are collected.
- These images are used as data.
- The AI system learns facial features from these images.

Therefore, images act as data for training the AI system.

Remember

Data is the fuel of an AI system.

Without fuel, a vehicle cannot move.

Similarly, without data, AI cannot learn.

1.3.2 Model

What is a Model?

A Model is the intelligent part or "brain" of an AI system.

It learns patterns from data and uses those patterns to make predictions, recommendations, or decisions.

Simple Definition

A Model is a trained AI program that learns from data and makes predictions.

How a Model Works

Step 1

The model receives data.

Step 2

It learns patterns from the data.

Step 3

It uses those patterns to make decisions or predictions.

Examples of AI Models

- Machine Learning Models
- Deep Learning Models

- Large Language Models (LLMs)
 - Image Recognition Models
 - Speech Recognition Models
-

Real-Life Example

ChatGPT uses a Large Language Model (LLM).

The model learns from a large amount of text and generates meaningful answers to user questions.

Remember

Model is the brain of the AI system.

Data teaches the model.

The model makes decisions.

1.3.3 Interface

What is an Interface?

The Interface is the part through which users interact with an AI system.

It acts as a bridge between the user and the AI model.

Simple Definition

Interface is the medium through which users communicate with an AI system.

Types of Interfaces

Web Interface

Example: ChatGPT website

Mobile Application

Example: Google Maps app

Chatbot

Example: Customer support chatbot

Voice Assistant

Example: Siri or Google Assistant

Dashboard

Example: Business analytics dashboard

Importance of Interface

A good interface:

- Is easy to use
 - Displays information clearly
 - Improves user experience
 - Helps users achieve their goals quickly
-

Real-Life Example

When you type a question into ChatGPT, the text box, buttons, and chat screen form the interface.

The interface sends your question to the AI model and displays the response.

Remember

Interface is the face of the AI system.

Users interact with the interface, not directly with the AI model.

1.3.4 Feedback Loop

What is a Feedback Loop?

A Feedback Loop is a process through which an AI system receives feedback about its performance and uses that feedback to improve itself.

Simple Definition

A Feedback Loop collects user responses and uses them to improve the AI system.

How Feedback Loop Works

Step 1

AI generates a result.

Step 2

Users provide feedback.

Step 3

The system analyzes the feedback.

Step 4

The model is improved using the feedback.

Step 5

The AI provides better results in the future.

Real-Life Example

In YouTube:

- You watch videos.
- You like, dislike, or skip videos.
- YouTube records this feedback.
- The recommendation system improves future suggestions.

This process is called a feedback loop.

Benefits of Feedback Loop

- Improves accuracy
 - Enhances user experience
 - Reduces errors
 - Helps AI learn continuously
 - Keeps the system updated
-

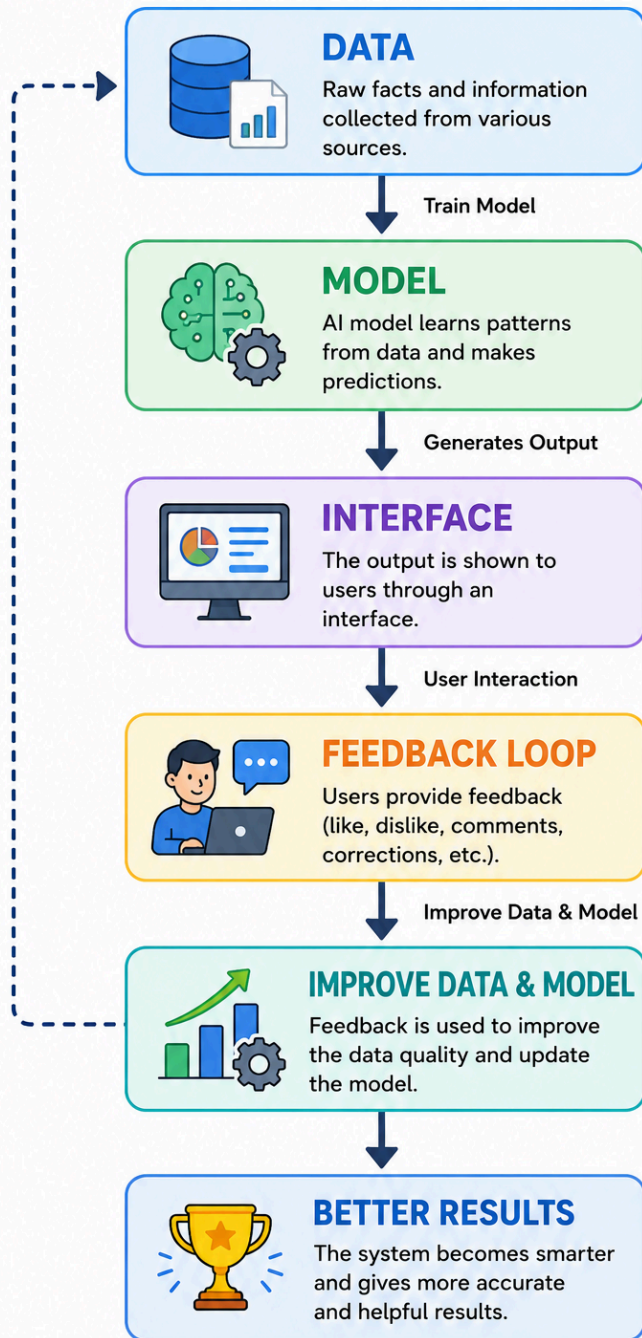
Remember

Feedback Loop is the teacher of the AI system.

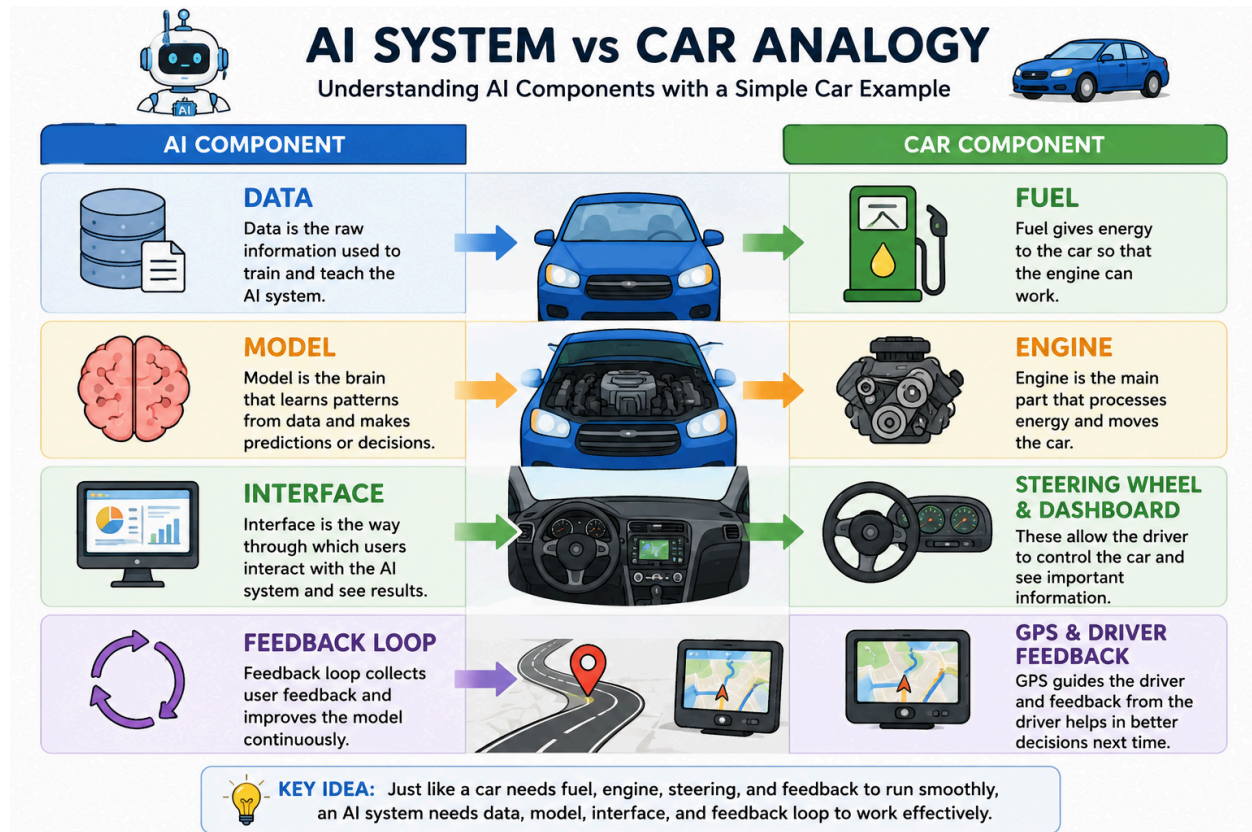
It tells the AI what is correct and what needs improvement.

Complete AI System Flow

HOW AN AI SYSTEM WORKS



Easy Analogy: AI System as a Car



1.4 Overview of Generative AI

What is Generative AI?

Generative AI is a type of Artificial Intelligence that can create new content such as text, images, audio, videos, and computer code.

Unlike traditional AI, which mainly analyzes data and makes predictions, Generative AI can generate new content that did not previously exist.

Simple Definition

Generative AI is AI that creates new content based on the data it has learned.

Examples of Generative AI

Text Generation

- ChatGPT
- Gemini
- Claude

These tools can write:

- Essays
- Emails
- Stories
- Computer programs

Image Generation

- DALL·E
- Midjourney
- Adobe Firefly

These tools can create images from text descriptions.

Music Generation

AI can compose songs and background music.

Video Generation

AI can generate videos from text instructions.

How Generative AI Works

Step 1: Training

The AI learns from a large amount of data.

Example:

- Books
- Articles
- Images
- Audio recordings

Step 2: Understanding Patterns

The AI identifies patterns and relationships in the data.

Step 3: Content Generation

When a user provides a prompt, the AI generates new content based on its learning.

Example

User Prompt:

"Write a short story about a robot."

Generative AI creates a completely new story.

Applications of Generative AI

Education

- Notes preparation
- Question generation
- Learning assistance

Business

- Email writing
- Report generation
- Customer support

Design

- Logo creation
- Poster design
- Image generation

Programming

- Code generation
 - Bug fixing
 - Documentation writing
-

Advantages of Generative AI

- ✓ Saves time
 - ✓ Increases productivity
 - ✓ Encourages creativity
 - ✓ Automates repetitive work
 - ✓ Helps generate ideas
-

Limitations of Generative AI

- ✗ Can provide incorrect information
- ✗ May generate biased content

- ✗ Requires quality data
 - ✗ Cannot replace human judgment completely
-

Remember

Generative AI = Create New Content

Examples:

- ChatGPT → Text
 - DALL·E → Images
 - Suno → Music
 - Sora → Videos
-

1.5 Introduction to Large Language Models (LLMs)

What is an LLM?

LLM stands for **Large Language Model**.

It is a special type of AI model trained on a huge amount of text data to understand and generate human language.

Simple Definition

An LLM is an AI model that can read, understand, and generate human language.

Why is it called "Large"?

Because it is trained on:

- Millions of books
- Articles

- Websites
- Documents
- Conversations

The word "Large" refers to the huge amount of data and parameters used during training.

Examples of LLMs

- ChatGPT
 - GPT Models
 - Gemini
 - Claude
 - Llama
-

What Can LLMs Do?

Answer Questions

Example:

"What is Artificial Intelligence?"

Generate Text

Example:

Write an essay on climate change.

Summarize Information

Convert long text into short notes.

Translate Languages

English ↔ Gujarati

English ↔ Hindi

Generate Code

Create programs in Python, Java, C++, etc.

How LLMs Work

Input

User enters a prompt.

Example:

"Explain AI in simple words."

Processing

The LLM analyzes the text and predicts the most suitable response.

Output

A meaningful answer is generated.

Example

Prompt:

"What is the capital of India?"

Output:

"New Delhi"

Real-Life Example

Think of an LLM as a very knowledgeable digital assistant.

It has read a huge amount of information and can answer questions quickly.

Remember

LLM = AI Brain for Language

Input → Understand → Generate Response

1.6 Types of AI Models (Conceptual Overview)

AI models can be classified according to the type of information they process.

The four common types are:

1. Text Models
 2. Image Models
 3. Speech Models
 4. Embedding Models
-

1.6.1 Text Models

What are Text Models?

Text models process and generate text.

They understand written language and produce text-based outputs.

Examples

- ChatGPT
- Gemini
- Claude

Uses

- Question answering
- Essay writing
- Translation
- Summarization

- Chatbots

Input and Output

Input: Text

Output: Text

Example

Input:

"Explain AI."

Output:

Artificial Intelligence is a technology that enables machines to perform intelligent tasks.

1.6.2 Image Models

What are Image Models?

Image models process, analyze, and generate images.

Examples

- DALL·E
- Midjourney
- Stable Diffusion

Uses

- Image generation
- Face recognition
- Medical image analysis
- Object detection

Input and Output

Input: Text or Image

Output: Image

Example

Input:

"Create an image of a smart robot."

Output:

A newly generated robot image.

1.6.3 Speech Models

What are Speech Models?

Speech models process and generate human speech.

Uses

- Voice assistants
- Speech recognition
- Voice translation
- Audio transcription

Examples

- Siri
- Google Assistant
- Alexa

Input and Output

Input: Voice

Output: Text or Voice

Example

Input:

"Call my friend."

Output:

The assistant performs the action.

1.6.4 Embedding Models

What are Embedding Models?

Embedding models convert text, images, or other data into numerical representations called vectors.

These vectors help computers understand similarity and relationships between data.

Simple Definition

Embedding models convert information into numbers that AI can understand.

Uses of Embedding Models

Search Systems

Finding relevant documents.

Recommendation Systems

Suggesting products or movies.

Semantic Search

Finding information based on meaning.

Similarity Detection

Comparing documents and images.

Example

Search Query:

"Best AI books"

The embedding model finds books with similar meaning, even if the exact words are different.

Easy Analogy

Imagine a library.

- Text Models = Writer
- Image Models = Artist
- Speech Models = Speaker and Listener
- Embedding Models = Librarian who organizes information